13 Extra laboratory 3

13. Visual Odometry

13.1 Visual Odometry

Visual Odometry is defined as the process of estimating the robot's motion (translation and rotation with respect to a reference frame) by observing a sequence of images of its environment. VO is a particular case of a technique known as Structure From Motion (SFM) that tackles the problem of 3D reconstruction of both the structure of the environment and camera poses from sequentially ordered or unordered image sets. SFM's final refinement and global optimisation step of both the camera poses and the structure is computationally expensive and usually performed off-line. However, the estimation of the camera poses in VO is required to be conducted in real-time. In recent years, many VO methods have been proposed which those can be divided into monocular and stereo camera methods. These methods are then further divided into feature matching (matching features over a number of frames), feature tracking (matching features in adjacent frames) and optical flow techniques (based on the intensity of all pixels or specific regions in sequential images).

Simultaneous localisation and mapping is an extension of the VO method that deals with drift, but also SLAM is a process in which a robot is required to localise itself in an unknown environment and build a map of this environment at the same time without any prior information with the aid of external sensors (or a single sensor).

Visual odometry consists of several steps:

1. Feature detection (usually ORB – FAST+BRIEF)
2. Feature matching (Brute-force or FLANN)
3. 3D point cloud generation – triangulation
4. Inliers detection, RANSAC filtering
5. Motion estimation – determination of rotation and translation

13.2 Object tracking - exercise

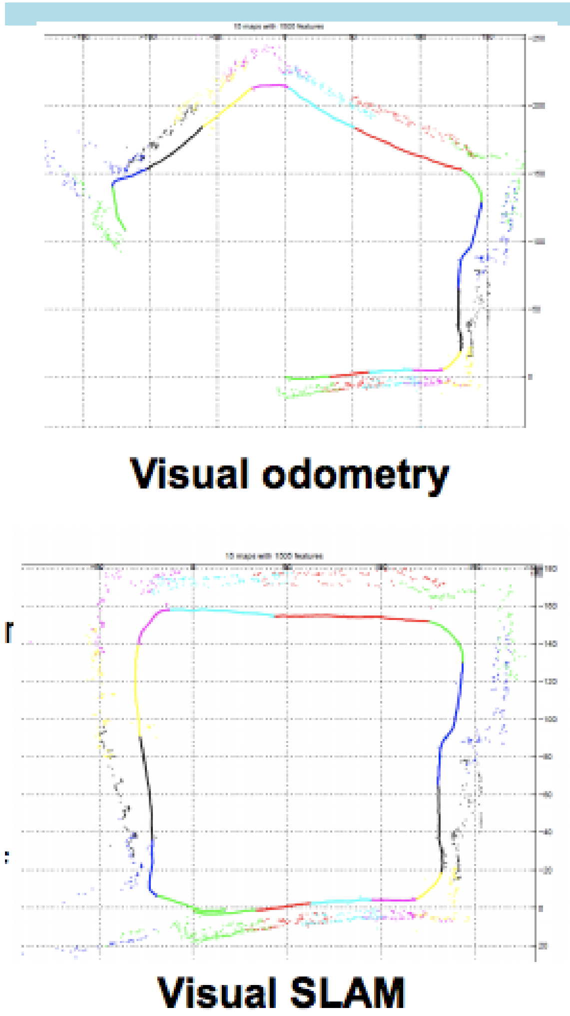

Figure 13.1: Comparison of VO and SLAM.

Exercise 13.1 The purpose of the exercise is to implement simple visual odometry, i.e. to determine the trajectory of a car moving through the streets on the basis of a KITTI dataset.

1. Download the materials and the exercise template from the UPeL platform.
2. The exercise has been prepared in Jupyter notebook. The file `.ipynb` can be opened and run using Google Colab (www.colab.research.google.com). It is also possible to run the file using the VS Code editor. You will probably be required to install the appropriate libraries. If we choose VS Code, we skip loading the library:

```
from google.colab import drive
drive.mount('/content/drive')
```

3. Follow the instructions in the file.
4. The solution to the task is the completed and working .ipynb file, which should be sent to the UPeL platform.